

Naive Bayes Classifier

Machine Learning: Comprehensive Lecture Notes

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February 10, 2026

 Machine Learning Series 

Abstract

This comprehensive lecture note covers the Naive Bayes Classifier, a fundamental probabilistic machine learning algorithm based on Bayes' Theorem. The notes include theoretical foundations, mathematical derivations, multiple worked examples (Tennis Playing, Medical Diagnosis, Text Classification), and practical applications. Each section includes detailed explanations, step-by-step solutions, and mathematical formulations. The document concludes with practice problems and solutions to reinforce understanding.

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1 Introduction to Naive Bayes Classifier

Theory Box

Naive Bayes is a supervised machine learning algorithm primarily used for classification problems. It's called "naive" because it makes a strong assumption that all features are conditionally independent given the class label. Despite this simplification, it performs remarkably well in many real-world applications.

1.1 Key Characteristics

- ▶ **Probabilistic Model:** Based on Bayes' Theorem with conditional independence assumptions
- ▶ **Supervised Learning:** Requires labeled training data
- ▶ **Classification Focus:** Predicts discrete class labels
- ▶ **Efficient Computation:** Simple and fast, even with large datasets
- ▶ **Baseline Algorithm:** Often used as a benchmark for text classification

1.2 Common Applications

Application	Example Use Case
Spam Filtering	Classifying emails as spam/ham
Sentiment Analysis	Determining positive/negative reviews
Document Classification	Categorizing news articles
Medical Diagnosis	Predicting diseases from symptoms
Recommendation Systems	User behavior classification

2 Bayes' Theorem Foundations

2.1 The Original Formula

The fundamental equation governing Naive Bayes is:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad (1)$$

Term	Description
$P(A B)$	Posterior probability of A given evidence B
$P(B A)$	Likelihood of evidence B given A is true
$P(A)$	Prior probability of hypothesis A
$P(B)$	Marginal probability of evidence B

2.2 Example: Total Probability with Two Bags

Example

Problem: Two bags contain different colored balls:

- **Bag 1 (B1):** 2 Red and 3 Black balls (Total: 5)
- **Bag 2 (B2):** 4 Red and 3 Black balls (Total: 7)

If we randomly choose one bag and then draw one ball, what is the total probability of drawing a Red ball?

Solution:

1. **Prior probabilities:**

$$P(B1) = \frac{1}{2}, \quad P(B2) = \frac{1}{2}$$

2. **Likelihoods:**

$$P(\text{Red} | B1) = \frac{2}{5}, \quad P(\text{Red} | B2) = \frac{4}{7}$$

3. **Total probability of Red:**

$$\begin{aligned} P(\text{Red}) &= [P(B1) \times P(\text{Red} | B1)] + [P(B2) \times P(\text{Red} | B2)] \\ &= \left(\frac{1}{2} \times \frac{2}{5}\right) + \left(\frac{1}{2} \times \frac{4}{7}\right) \\ &= \frac{1}{5} + \frac{2}{7} = \frac{7+10}{35} = \frac{17}{35} \end{aligned}$$

2.3 Example: Applying Bayes' Theorem (Reverse Probability)

Example

Problem: We drew a ball and it was **Red**. What is the probability it came from **Bag 1**?

Solution: Using Bayes' Theorem:

$$P(B1 | \text{Red}) = \frac{P(\text{Red} | B1) \cdot P(B1)}{P(\text{Red})}$$

1. **Numerator:**

$$P(\text{Red} | B1) \times P(B1) = \frac{2}{5} \times \frac{1}{2} = \frac{1}{5}$$

2. **Denominator:** From previous example, $P(\text{Red}) = \frac{17}{35}$

3. **Final calculation:**

$$P(B1 | \text{Red}) = \frac{1/5}{17/35} = \frac{1}{5} \times \frac{35}{17} = \frac{7}{17} \approx 0.41$$

So, there is a 41% chance the red ball came from Bag 1.

3 Adaptation for Classification Problems

3.1 Problem Formulation

For a dataset with n features and class variable Y :

- Features: $X_1, X_2, \dots, X_n$
- Target: Y (e.g., Yes/No, Positive/Negative)
- Goal: Find $P(Y | X_1, X_2, \dots, X_n)$

3.2 Derivation Process

Starting from Bayes' Theorem:

$$P(Y | X_1, \dots, X_n) = \frac{P(X_1, \dots, X_n | Y) \cdot P(Y)}{P(X_1, \dots, X_n)} \quad (2)$$

$$\propto P(X_1, \dots, X_n | Y) \cdot P(Y) \quad (3)$$

3.3 The "Naive" Assumption

Important Note

Conditional Independence Assumption: The Naive Bayes classifier assumes that all features are independent of one another, given the class. This means that the presence of one feature does not affect the presence of another feature within the same class.

Mathematically:

$$P(X_1, \dots, X_n | Y) = \prod_{i=1}^n P(X_i | Y)$$

This is the "naive" part of Naive Bayes, as real-world features often have dependencies. However, this simplification makes the math tractable and the algorithm fast and effective.

3.4 Final Classification Formula

$$P(Y | X_1, \dots, X_n) \propto P(Y) \cdot \prod_{i=1}^n P(X_i | Y) \quad (4)$$

3.5 Decision Rule: ArgMax

The predicted class $\hat{Y}$ is:

$$\hat{Y} = \underset{Y}{\operatorname{argmax}} \left[P(Y) \cdot \prod_{i=1}^n P(X_i | Y) \right] \quad (5)$$

The denominator $P(X_1, \dots, X_n)$ is dropped because it is constant for all classes and thus does not affect which class has the maximum probability.

4 Complete Example: Tennis Playing Decision

4.1 Dataset Overview

Outlook	Temperature	Humidity	Play Tennis?
Sunny	Hot	High	No
Sunny	Hot	High	No
Overcast	Hot	High	Yes
Rainy	Mild	High	Yes
Rainy	Cool	Normal	Yes
Rainy	Cool	Normal	No
Overcast	Cool	Normal	Yes
Sunny	Mild	High	No
Sunny	Cool	Normal	Yes
Rainy	Mild	Normal	Yes
Sunny	Mild	Normal	Yes
Overcast	Mild	High	Yes
Overcast	Hot	Normal	Yes
Rainy	Mild	High	No

Table 1: Weather conditions and tennis playing decisions

4.2 Probability Calculations

4.2.1 Prior Probabilities

$$P(Yes) = \frac{9}{14} \approx 0.643$$

$$P(No) = \frac{5}{14} \approx 0.357$$

4.2.2 Conditional Probabilities

$$P(Sunny | Yes) = \frac{2}{9} \approx 0.222$$

$$P(Sunny | No) = \frac{3}{5} = 0.600$$

$$P(Hot | Yes) = \frac{2}{9} \approx 0.222$$

$$P(Hot | No) = \frac{2}{5} = 0.400$$

$$P(High | Yes) = \frac{3}{9} \approx 0.333$$

$$P(High | No) = \frac{4}{5} = 0.800$$

4.3 Step-by-Step Classification

Example

Scenario: Outlook = Sunny, Temperature = Hot, Humidity = High
Should we play tennis?

Solution:

For **Yes** class:

$$\begin{aligned}P(Yes | Sunny, Hot, High) &\propto P(Sunny | Yes) \cdot P(Hot | Yes) \cdot P(High | Yes) \cdot P(Yes) \\&= \frac{2}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{9}{14} \\&= \frac{12}{729} \times \frac{9}{14} \approx 0.0106\end{aligned}$$

For **No** class:

$$\begin{aligned}P(No | Sunny, Hot, High) &\propto P(Sunny | No) \cdot P(Hot | No) \cdot P(High | No) \cdot P(No) \\&= \frac{3}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{5}{14} \\&= \frac{24}{125} \times \frac{5}{14} \approx 0.0686\end{aligned}$$

Normalization:

$$\begin{aligned}\text{Total} &= 0.0106 + 0.0686 = 0.0792 \\P(Yes) &= \frac{0.0106}{0.0792} \approx 0.134 \\P(No) &= \frac{0.0686}{0.0792} \approx 0.866\end{aligned}$$

Decision: Since $P(No) > P(Yes)$, the prediction is **Do Not Play Tennis**.

5 Medical Diagnosis Example

5.1 Dataset Overview

Person	COVID	Flu	Fever (Class)
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	Yes	Yes
7	No	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	Yes	No	No

Table 2: Patient medical data for 10 patients

5.2 Probability Calculations

5.2.1 Prior Probabilities

$$P(Fever = Yes) = \frac{7}{10} = 0.7$$
$$P(Fever = No) = \frac{3}{10} = 0.3$$

5.2.2 Likelihood Tables

Likelihood	Fever = Yes	Fever = No
COVID=Yes	4/7	2/3
COVID=No	3/7	1/3
Flu=Yes	3/7	2/3
Flu=No	4/7	1/3

Table 3: Conditional probabilities for medical diagnosis

5.3 Step-by-Step Classification

Example

Problem: A new patient has **Flu = Yes** and **COVID = No**.

Should we predict Fever = Yes or Fever = No?

Solution:

For **Fever=Yes**:

$$\begin{aligned}P(Yes | Flu = Yes, COVID = No) &\propto P(Flu = Yes | Yes) \times P(COVID = No | Yes) \times P(Yes) \\&= \frac{3}{7} \times \frac{3}{7} \times \frac{7}{10} \\&= \frac{9}{49} \times \frac{7}{10} = \frac{63}{490} \approx 0.1286\end{aligned}$$

For **Fever=No**:

$$\begin{aligned}P(No | Flu = Yes, COVID = No) &\propto P(Flu = Yes | No) \times P(COVID = No | No) \times P(No) \\&= \frac{2}{3} \times \frac{1}{3} \times \frac{3}{10} \\&= \frac{2}{9} \times \frac{3}{10} = \frac{6}{90} \approx 0.0667\end{aligned}$$

Normalization:

$$\begin{aligned}\text{Total} &= 0.1286 + 0.0667 = 0.1953 \\P(Yes) &= \frac{0.1286}{0.1953} \approx 0.658 \\P(No) &= \frac{0.0667}{0.1953} \approx 0.342\end{aligned}$$

Decision: Since $P(Yes) > P(No)$, the prediction is **Fever = Yes**.

6 Text Classification with Naive Bayes

6.1 NLP Pipeline for Text Classification

1. **Tokenization:** Split text into words/tokens
2. **Stopword Removal:** Remove common words (the, is, and, etc.)
3. **Stemming/Lemmatization:** Reduce words to root form
4. **Vectorization:** Convert text to numerical features
 - Bag of Words (BoW)
 - TF-IDF (Term Frequency-Inverse Document Frequency)

6.2 Sample Dataset for Sentiment Analysis

Sentence	the	food	delicious	bad	Sentiment
The food is delicious	1	1	1	0	Positive (1)
The food is bad	1	1	0	1	Negative (0)
Food delicious	0	1	1	0	Positive (1)
The bad food	1	1	0	1	Negative (0)
Delicious	0	0	1	0	Positive (1)

6.3 Probability Calculations for Text

$$P(\text{Positive}) = \frac{3}{5} = 0.6$$

$$P(\text{Negative}) = \frac{2}{5} = 0.4$$

6.4 Zero Probability Problem and Solution

Important Note

Problem: If a word doesn't appear in a class, $P(\text{word} | \text{class}) = 0$, making the entire product zero.

Solution: Laplace Smoothing (Add-one smoothing):

$$P(\text{word} | \text{class}) = \frac{\text{count}(\text{word in class}) + 1}{\text{total words in class} + V}$$

Where V is the vocabulary size.

6.4.1 Word Probabilities (with Laplace smoothing, $\alpha = 1$)

With vocabulary size $V = 4$ (the, food, delicious, bad):

$$P(\text{the} | \text{Positive}) = \frac{1 + 1}{10 + 4} = \frac{2}{14} \approx 0.143$$

$$P(\text{food} | \text{Positive}) = \frac{2 + 1}{10 + 4} = \frac{3}{14} \approx 0.214$$

$$P(\text{delicious} | \text{Positive}) = \frac{3 + 1}{10 + 4} = \frac{4}{14} \approx 0.286$$

$$P(\text{bad} | \text{Positive}) = \frac{0 + 1}{10 + 4} = \frac{1}{14} \approx 0.071$$

7 Advantages and Limitations

7.1 Advantages

Advantage	Explanation
Simple and Fast	Computationally efficient, works well with high-dimensional data
Requires Less Data	Can perform well even with small datasets
Handles Multiple Classes	Naturally extends to multi-class problems
Works with Categorical Features	Ideal for text and categorical data
Robust to Irrelevant Features	Independence assumption helps ignore irrelevant features
Probabilistic Outputs	Provides probability estimates for predictions

7.2 Limitations

Limitation	Explanation
Naive Independence Assumption	Features are often correlated in real-world data
Zero Frequency Problem	Requires smoothing techniques for unseen features
Bad Probability Estimator	Probabilities are not always well-calibrated
Sensitive to Feature Selection	Performance depends on relevant feature selection
Assumes Feature Distribution	Need to choose appropriate variant for data type

8 Naive Bayes Variants

8.1 Gaussian Naive Bayes

- **For:** Continuous features
- **Assumption:** Features follow normal distribution

- **Likelihood:** $P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$

8.2 Multinomial Naive Bayes

- **For:** Discrete counts (word counts in text)
- **Assumption:** Features represent frequency counts

- **Likelihood:** $P(x_i | y) = \frac{N_{yi} + \alpha}{N_y + \alpha n}$

8.3 Bernoulli Naive Bayes

- ▶ **For:** Binary/boolean features
- ▶ **Assumption:** Features are independent boolean variables
- ▶ **Likelihood:** $P(x_i | y) = P(i | y) x_i + (1 - P(i | y))(1 - x_i)$

8.4 Categorical Naive Bayes

- ▶ **For:** Categorical features with multiple categories
- ▶ **Assumption:** Each feature has categorical distribution
- ▶ **Likelihood:** $P(x_i | y) = \frac{N_{yik} + \alpha}{N_y + \alpha n_i}$

9 Practice Problems

Example

Problem 1: Medical Diagnosis

A hospital uses symptoms to diagnose Disease D:

- $P(D) = 0.01$ (1% of patients have the disease)
- $P(\text{Fever} | D) = 0.9$
- $P(\text{Cough} | D) = 0.8$
- $P(\text{Fever} | \neg D) = 0.1$
- $P(\text{Cough} | \neg D) = 0.2$

A patient presents with both fever and cough. What is the probability they have Disease D?

Example

Problem 2: Email Spam Classification

From 1000 emails:

- 300 are spam, 700 are ham
- Word "free" appears in 250 spam and 50 ham emails
- Word "money" appears in 200 spam and 30 ham emails

Calculate $P(\text{Spam} | \text{"free money"})$ and $P(\text{Ham} | \text{"free money"})$ for a new email containing both words.

Example

Problem 3: Weather Prediction

Using the tennis dataset:

- Outlook = Rainy
- Temperature = Cool
- Humidity = Normal

Calculate whether to play tennis or not.

Example

Problem 4: Fruit Identification

A model is trained to identify fruits based on Color and Shape. Predict the fruit for a new sample that is '(Color=Red, Shape=Round)'.

Priors: $P(\text{Apple}) = 0.5$, $P(\text{Grape}) = 0.5$.

Likelihoods:

$$P(\text{Red} | \text{Apple}) = 0.8$$

$$P(\text{Round} | \text{Apple}) = 0.9$$

$$P(\text{Red} | \text{Grape}) = 0.3$$

$$P(\text{Round} | \text{Grape}) = 0.1$$

Example

Problem 5: Multi-class Classification

A document classifier has 3 categories:

- $P(\text{Sports}) = 0.4$, $P(\text{Politics}) = 0.3$, $P(\text{Technology}) = 0.3$
- For words "game" and "vote":

$$P(\text{game} | \text{Sports}) = 0.6$$

$$P(\text{vote} | \text{Sports}) = 0.1$$

$$P(\text{game} | \text{Politics}) = 0.1$$

$$P(\text{vote} | \text{Politics}) = 0.7$$

$$P(\text{game} | \text{Tech}) = 0.05$$

$$P(\text{vote} | \text{Tech}) = 0.2$$

Classify a document containing both "game" and "vote".

10 Solutions

10.1 Solution 1: Medical Diagnosis

$$\begin{aligned}P(D | F, C) &\propto P(F | D) \times P(C | D) \times P(D) \\&= 0.9 \times 0.8 \times 0.01 = 0.0072 \\P(\neg D | F, C) &\propto P(F | \neg D) \times P(C | \neg D) \times P(\neg D) \\&= 0.1 \times 0.2 \times 0.99 = 0.0198 \\ \text{Total} &= 0.0072 + 0.0198 = 0.027 \\P(D) &= \frac{0.0072}{0.027} \approx 0.267 \\P(\neg D) &= \frac{0.0198}{0.027} \approx 0.733\end{aligned}$$

Answer: Approximately 26.7% chance of having Disease D.

10.2 Solution 2: Email Spam Classification

$$\begin{aligned}P(\text{Spam}) &= \frac{300}{1000} = 0.3 \\P(\text{Ham}) &= \frac{700}{1000} = 0.7 \\P(\text{free} | \text{Spam}) &= \frac{250}{300} \approx 0.833 \\P(\text{money} | \text{Spam}) &= \frac{200}{300} \approx 0.667 \\P(\text{free} | \text{Ham}) &= \frac{50}{700} \approx 0.0714 \\P(\text{money} | \text{Ham}) &= \frac{30}{700} \approx 0.0429 \\P(\text{Spam} | \text{"free money"}) &\propto 0.3 \times 0.833 \times 0.667 \approx 0.1667 \\P(\text{Ham} | \text{"free money"}) &\propto 0.7 \times 0.0714 \times 0.0429 \approx 0.0021 \\ \text{Total} &= 0.1667 + 0.0021 = 0.1688 \\P(\text{Spam}) &= \frac{0.1667}{0.1688} \approx 0.987 \\P(\text{Ham}) &= \frac{0.0021}{0.1688} \approx 0.013\end{aligned}$$

Answer: Email is almost certainly spam (98.7% probability).

10.3 Solution 3: Weather Prediction

From tennis dataset:

$$P(Rainy | Yes) = \frac{3}{9} \approx 0.333$$

$$P(Cool | Yes) = \frac{3}{9} \approx 0.333$$

$$P(Normal | Yes) = \frac{6}{9} \approx 0.667$$

$$P(Rainy | No) = \frac{2}{5} = 0.4$$

$$P(Cool | No) = \frac{1}{5} = 0.2$$

$$P(Normal | No) = \frac{1}{5} = 0.2$$

$$P(Yes | Rainy, Cool, Normal) \propto 0.333 \times 0.333 \times 0.667 \times 0.643 \approx 0.0475$$

$$P(No | Rainy, Cool, Normal) \propto 0.4 \times 0.2 \times 0.2 \times 0.357 \approx 0.0057$$

$$\text{Total} = 0.0475 + 0.0057 = 0.0532$$

$$P(Yes) = \frac{0.0475}{0.0532} \approx 0.893$$

$$P(No) = \frac{0.0057}{0.0532} \approx 0.107$$

Answer: Play tennis (89.3% probability of Yes).

10.4 Solution 4: Fruit Identification

$$\begin{aligned} P(Apple | Red, Round) &\propto P(Red | Apple) \times P(Round | Apple) \times P(Apple) \\ &= 0.8 \times 0.9 \times 0.5 = 0.36 \end{aligned}$$

$$\begin{aligned} P(Grape | Red, Round) &\propto P(Red | Grape) \times P(Round | Grape) \times P(Grape) \\ &= 0.3 \times 0.1 \times 0.5 = 0.015 \end{aligned}$$

$$\text{Total} = 0.36 + 0.015 = 0.375$$

$$P(Apple) = \frac{0.36}{0.375} = 0.96$$

$$P(Grape) = \frac{0.015}{0.375} = 0.04$$

Answer: Apple (96% probability).

10.5 Solution 5: Multi-class Classification

$$P(\text{Sports} | \text{game}, \text{vote}) \propto 0.4 \times 0.6 \times 0.1 = 0.024$$

$$P(\text{Politics} | \text{game}, \text{vote}) \propto 0.3 \times 0.1 \times 0.7 = 0.021$$

$$P(\text{Tech} | \text{game}, \text{vote}) \propto 0.3 \times 0.05 \times 0.2 = 0.003$$

$$\text{Total} = 0.024 + 0.021 + 0.003 = 0.048$$

$$P(\text{Sports}) = \frac{0.024}{0.048} = 0.500$$

$$P(\text{Politics}) = \frac{0.021}{0.048} = 0.438$$

$$P(\text{Tech}) = \frac{0.003}{0.048} = 0.062$$

Answer: Sports category (50% probability).

11 Summary and Key Points

☰ Summary & Key Takeaways

Core Principles of Naive Bayes:

1. **Foundation:** Based on Bayes' Theorem: $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$
2. **The "Naive" Part:** Assumes all features are conditionally independent given the class: $P(X_1, \dots, X_n | Y) = \prod P(X_i | Y)$
3. **Classification Rule:** Uses maximum a posteriori (MAP) estimation: $\hat{Y} = \operatorname{argmax}_Y P(Y) \prod P(X_i | Y)$
4. **Training:** Requires calculating prior probabilities and conditional probabilities from training data
5. **Smoothing:** Often needs Laplace smoothing to handle unseen features

11.1 Key Formulas

1. Bayes' Theorem: $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$
2. Naive Bayes: $P(Y | X) \propto P(Y) \prod P(X_i | Y)$
3. Decision Rule: $\hat{Y} = \operatorname{argmax}_Y P(Y) \prod P(X_i | Y)$
4. Laplace Smoothing: $\frac{\text{count} + \alpha}{\text{total} + \alpha V}$

11.2 When to Use Naive Bayes

- **Recommended:** Text classification, categorical data, quick prototypes, high-dimensional data
- **Not Recommended:** When feature dependencies are critical, probability calibration needed
- **Consider:** As a baseline model before trying more complex algorithms